22PEPB08 – Industrial load Modelling and Controlly

REG.NO: | | | | | | | | | | | |

GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – Nov. / Dec. 2023

Fourth Semester

ELECTRONICS and COMMUNICATION ENGINEERING

18LPC404 – TRANSMISSION LINES and WAVE GUIDES

Regulations 2018

Time : 3 Hours | Maximum Marks : 100

Answer ALL Questions

Part - A **(10 x 2 = 20 Marks)**

No.	Question	M	CO	BT
1.	Define velocity of propagation.	2	CO1	R
2.	What are the properties of infinite length?	2	CO1	R
3.	Define nodes and antinodes.	2	CO2	R
4.	Write the expression for SWR in terms of reflection coefficient.	2	CO2	R
5.	Give the expression for characteristic impedance of the quarter wave transform.	2	CO3	R
6.	What are the disadvantages of single stub matching?	2	CO3	R
7.	Write the characteristics of TEM waves?	2	CO4	U
8.	Define guided waves.	2	CO4	R
9.	Mention the application of circular waveguide.	2	CO5	R
10.	Define the quality of a cavity resonator.	2	CO5	R

Part – B **(5 x 16 = 80 Marks)**

No.			Question	M	CO	BT
11.	a)		Derive the conditions required for a distortion less line.	16	CO1	AN
			(OR)			
	b)		Derive the general transmission line equations for voltage and current at any point on a line.	16	CO1	U
12.	a)	i)	Discuss the theory of open and short circuited lines with voltage and current distribution diagrams and also get the input impedance.	8	CO2	AN
		ii)	A line having a characteristic impedance of 50Ω is terminated in load impedance (75+j75)Ω. Determine the reflection coefficient and VSWR.	8	CO2	AN
			(OR)			
	b)	i)	With neat diagram, briefly explain the standing waves.	8	CO2	A
		ii)	A radio frequency line with Z_0=70Ω is terminated by Z_L=115-j80Ω at λ=2.5m. Find VSWR and the minimum and maximum line impedances.	8	CO2	AN

13.	a)	A single stub is to match a 400Ω line to a load of 200-j100Ω. The wavelength is 3m. Determine the position and length of the short circuit	16	CO3	A
		(OR)			
	b)	Discuss the operation of quarter wave transformer and also design a quarter wave transformer to match a load of 200Ω to a source resistance of 500Ω, operating at a frequency of 200MHz.	16	CO3	U
14.	a)	A pair of perfectly conducting plates are separated by 3 cm in air and carries a 10GHz signal in TM_1 mode. Find the cutoff frequency, phase constant and cutoff wavelength.	16	CO4	AN
		(OR)			
	b)	Explain the reasons for the attenuation of TE and TM waves between parallel plates with necessary expressions and diagrams.	16	CO4	AN
15.	a)	Discuss the field components of TE waves of rectangular waveguide.	16	CO5	U
		(OR)			
	b)	Explain the characteristics of TE waves in circular guides.	16	CO5	U

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BT	R	U	A	AN	E	C
%	10	36.6	13.3	31.1	-	-

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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

Full time B.E. Degree - Nov. / Dec. 2023 Examinations

Fourth Semester

ELECTRONICS AND COMMUNICATION ENGINEERING

20LPC404 – TRANSMISSION LINES and WAVE GUIDES

(Smith Chart is to be Provided)

Time : 3 Hours | Maximum Marks : 100

Answer ALL Questions

Part A **(10 x 2 = 20 Marks)**

1. What is an infinite line?
2. When will a transmission line deliver maximum power to the load?
3. Write the relation of SWR in terms of reflection coefficient.
4. What are nodes and antinodes on a line?
5. Distinguish between single stub and double stub matching.
6. What is the need for impedance matching?
7. What are the characteristics of ideal filters?
8. What is the significance of propagation constant in symmetrical network?
9. Why TEM waves are impossible in waveguide?
10. Give the expression for guided wavelength of rectangular waveguide.

Part B **(5 x 16 = 80 Marks)**

11. a) Obtain the general transmission line equation for the voltage and current at any point on a transmission line. (16)

(OR)

b) i) What is a loading? Specify the types of loading of lines. (8)

ii) Write a short note on reflection factor and reflection loss and give expressions. (8)

12. a) Explain how the VSWR and wavelength are measured over a transmission line. (16)

(OR)

b) i) Analyze the dissipation less line and derive the input impedance of the dissipation less line. (8)

ii) Illustrate the behaviour of the standing waves with neat diagram (8)

13. a) i) A load impedance of $90 - j50\ \Omega$ is to be matched to a line of $50\ \Omega$ using single stub matching. Find the length and position of the stub. (12)

ii) Design a Quarter wave transformer to match a load of 200Ω to a source resistance of 500Ω which operates at the frequency of 200 MHz. (4)

(OR)

b) i) Discuss in detail about the impedance matching by stubs. (4)

ii) A $300\ \Omega$ transmission line is connected to a load impedance of $(390+j600)\ \Omega$ at 10MHz.Evaluate the position and length of a short-circuited stub required to match the line using smith chart. (12)

14. a) What is m derived filter? Derive the relevant equation of m derived low pass filter. (16)

(OR)

b) Design a composite low pass filter with π-section to work into 600Ω with cutoff frequency of 2kHz and very high attenuation at 2.2 kHz. Also terminate filter properly. (16)

15. a) i) A pair of perfectly conducting plates are separated by 3 cm in air and carries a 10 GHz signal in TM 1 mode. Find the cutoff frequency, phase constant and cutoff wavelength. (8)

ii) Distinguish between the characteristics of TE and TM waves in rectangular waveguides. (8)

(OR)

b) i) Discuss briefly the attenuation of TE and TM waves between parallel plane. (12)

ii) A rectangular waveguide has the following dimensions a = 2.54 cm, b= 1.27 cm. Calculate the cut-off frequency for TE_{11} mode. (4)

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